Ihor Hrabar

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Summary

Calm and aspiring software developer with 3+ years of non-commercial software development experience. Able to develop solutions for a wide variety of areas, such as web front-ends, back-ends, writing business logic, working with low-level systems, etc. Currently studying at 42 School at Rome, and looking for a company to join and grow with.

Experience

Freelance — September 2018 to November 2018
Freelance Web Application Developer
Kyiv, Ukraine

- Developed a small proof-of-concept web application for an individual client.
- Studied Django in the process.

Komppomich — September 2020 to January 2021
Junior System Administrator
Kyiv, Ukraine

- Specialized in resolving technical, hardware-related issues.
- Communication with clients.
- Performed installation of various hardware at clients locations.

CGPA.Systems — June 2021 to September 2022
Junior Software Developer
Remote, Ukraine

- Worked extensively in C#.
- Specialized in writing business logic for input automation software.
- Bug fixing as an essential part of the work.

Goal2Be — December 2023 to December 2024
Back-end Developer
Remote, Italy

- Implemented content delivery micro-service for the Social Network.
- Integrated S3 storage solutions into the back-end.
- Worked in Kotlin.

Projects

A simple C library for 42 School works | *C*

- Written as a libft alternative. More on libft: https://github.com/vvarodi/libft_42#what-is-libft
- Written my own implementations of dynamic pointer arrays and hash maps data structures, strings and vectors,
- Fully compliant with the 42 Norm coding style.
- Project link: https://github.com/IgorGRBR/ylib

Top-down stealth-based puzzle game | *C, MiniLibX*

- Written as an implementation of so_long 42 project. Description of so_long project: https://github.com/deborabastos/42_so_long/blob/master/en.subject.pdf
- Developed a simple 2D maze puzzle game, where player has to reach the exit and avoid the patrolling guards.
- Implemented a simple 2D software renderer, capable of simple 2D scaling and translation transformations, and drawing textures to textures.
- Extended the default level format of the game while keeping it compatible with the usual so_long ".ber" format.
- Provided a robust asset-overloading system, allowing custom game graphics to be loaded from the levels.
- Supported on Linux, MacOS and WSL.
- Project link: https://github.com/IgorGRBR/so_sneaky

Unix shell re-implementation | *C*

- Written as an implementation of minishell 42 project. Description of minishell: https://github.com/JaeSeoKim/42cursus/blob/master/pdf/en.subject-Minishell.pdf
- Implemented a lexer-parser-interpreter pipeline for the execution of user-provided input.
- Built an interpreter based on a Abstract-Syntax-Tree walking technique.
- Implemented various Unix builtins, such as cd, echo, pwd, export, set, unset and env.
- Implemented "&&" and "||" operators with parenthesis support for grouping expressions.
- Implemented pipes with isolated environments and per-process threads using forks, allowing parallel execution of processes.
- Project link: https://github.com/IgorGRBR/lilshell

Raycasting-based First Person Shooter game engine | *C, SDL2, MiniLibX*

- Written as an implementation of cub3D 42 project. Description of cub3D project: https://github.com/frozenrobot/42-Cub3d/blob/master/cub3d-subject.pdf
- Extended software renderer from the so_long implementation, allowing for customizeable per-pixel effects to be applied during the rendering process, such as depth-based fog.
- Implemented raycasting with Digital Differential Analysis algorithm.
- Improved the rendering performance of images and scenes with multi-threading by utilizing thread pools.
- Designed the logic of enemy units with finite state machines and implemented path-finding with A* algorithm, allowing enemy NPCs to traverse complex scenes and pose a challenge to the player.
- Project link: https://github.com/IgorGRBR/cub3Demption

Low-battery shutdown daemon for Termux | *Rust, Termux-API*

- Written with Rust, using Termux-API to collect battery data and push notifications.
- Implemented a simple daemon that runs under Termux and automatically shuts down the device once a certain battery level threshold is reached.
- Daemon notifies the user if the battery level gets too close to the shutdown threshold with status bar notifications and text popups.
- Project link: https://github.com/IgorGRBR/lowbat-shutdown-daemon

Simple web server | *C++98*

- Written as an implementation of *webserv* from 42 School.
- Written with C++98, utilizing only std library.
- The web server features a decent range of functionality, and allows the user to define custom index files, custom error pages for various return codes, display directory listings, filter traffic based on its payload size or HTTP method type, etc.
- Implemented a simple configuration DSL, that allows the user to configure routes of the web server, as well as their behavior.
- Implemented IO multiplexing model to achieve higher throughput of the server, despite its single-threaded design.
- Implemented CGI support, verified with a few Python and Lua CGI scripts.
- Due to a few limitations of this project (such as mandatory use of C++98, and inability to use 3rd party libraries), I have implemented a hand-rolled implementation of certain C++03 features to allow safer memory management patterns and cleaner code appearance (namely shared/unique pointers and optional values).
- Achieved 100% availability rate when testing with siege.
- Project link: https://github.com/IgorGRBR/webserv

Skills

- High-level Linux development with C, C++, C#, Java, Python, Django.
- Back-end development with NestJS.
- Front-end development with React, NextJS, HTML, frontend for mobile, Kotlin.
- Experienced developer in JavaScript and TypeScript.
- Experienced in working with databases: SQL, MongoDB.
- Use of project and team management tools (Github, Mercurial) and issue trackers (GitHub, JIRA)
- Basic Agile familiarity
- Knowledge of imperative, object-oriented and functional programming paradigms, as well as various software development techniques and design patterns
- Optimal knowledge of several IDEs: Visual Studio Code, Visual Studio, Android Studio, JetBrains IntelliJ IDEA, Eclipse
- Usage experience of Windows & Unix-based systems
- Fast learner and a team player; really good at team leading.
- Excellent in problem analysis, fixing and optimizing.
- Excellent ability in code analyzing.
- Excellent verbal and written communication
- Experience of communicating and working with non-technical clients
- Native Ukrainian speaker with excellent knowledge of English. Good knowledge of Russian, basic understanding of Polish and Italian.

Education and Training

42 School. 2023 to Current
Software Engineering Bootcamp.
Rome, Italy.

Kyiv-Mohyla Academy. 2019 to 2021
Unfinished Masters degree in Computer Science
Kyiv, Ukraine.

Kyiv-Mohyla Academy. 2015 to 2019
Bachelor’s degree in Computer Science
Kyiv, Ukraine.